

Solving Simultaneous Equations

Task 1: Solve each simultaneous equation by **scaling** then **adding**.

Label each equation and solve simultaneously.

- | | |
|--------------------------------------|---------------------------------------|
| (a) $3x + 4y = 10$
$2x - 3y = 1$ | (b) $4x + 2y = 14$
$7x - 3y = 5$ |
| (c) $2x + 5y = 13$
$3x - 2y = 10$ | (d) $6x + 7y = 27$
$5x - 2y = -1$ |
| (e) $5x - 6y = -11$
$4x + 4y = 0$ | (f) $2x - 3y = -1$
$5x + 4y = -14$ |
| (g) $2x + 3y = 19$
$6x - 2y = -9$ | (h) $4x + 8y = 2$
$3x - 6y = -4.5$ |

Answers:

- a) $x = 2, y = 1$
- b) $x = 2, y = 3$
- c) $x = 4, y = 1$
- d) $x = 1, y = 3$
- e) $x = -1, y = 1$
- f) $x = -2, y = -1$
- g) $x = \frac{1}{2}, y = 6$
- h) $x = -\frac{1}{2}, y = \frac{1}{2}$

Task 2: Solve each by **scaling** then **deciding whether to add or subtract**.

Solve :-

- | | |
|---|--|
| (a) $3y + 2x = -10$
$2y + 5x = -14$ | (b) $8x - 8y = -16$
$5x - 3y = -28$ |
| (c) $6y + 6x = 0$
$7y + 5x = -1$ | (d) $8x + 6y = -8$
$2x + 11y = -2$ |
| (e) $3x = 2y - 5$
$4x - 3y + 8 = 0$ | (f) $2y = 5x + 16$
$3y - 5x - 19 = 0$ |
| (g) $2x - 3y - 13 = 0$
$3x + 2y = 0$ | (h) $2x + y - 5 = 0$
$3x - 4y - 13 = 0$ |

Answers:

- a) $x = -2, y = -2$
- b) $x = -11, y = -9$
- c) $x = \frac{1}{2}, y = -\frac{1}{2}$
- d) $x = -1, y = 0$
- e) $x = 1, y = 4$
- f) $x = -2, y = 3$
- g) $x = 2, y = -3$
- h) $x = 3, y = -1$